

THE CUBIC BORWEIN CONJECTURE: A COMPUTER-ASSISTED PROOF OF THE REMAINING RESIDUE CLASS

JAIDEEP SAI PADHI

ABSTRACT. Let $P_n(q) = \prod_{\substack{k \leq 3n, \\ 3 \nmid k}} (1 - q^k)$. The cubic Borwein conjecture asserts that the coefficients of $P_n(q)^3$ have the sign pattern $+$ $-$ $-$ according to the exponent modulo 3. Wang and Krattenthaler proved this for the exponents $\equiv 0, 1 \pmod{3}$. We prove the remaining class: $[q^m]P_n(q)^3 \leq 0$ for all $n \geq 1$ and all $m \equiv 2 \pmod{3}$.

This is a computer-assisted proof. The range $n \leq 243$ is an exact integer computation. For $n \geq 244$ the proof combines a band lemma, a Laplace-type integral representation, and a complex saddle-point analysis near the centre of the coefficient range. Every inequality on which the analytic part rests is verified in rigorous ball (interval) arithmetic, uniformly in n , by explicitly described certificates. The error term of a circle-method expansion that we use is proved in Appendix A. All programs, their expected outputs and the run commands are published with the paper.

1. INTRODUCTION

For $n \geq 1$ put

$$P_n(q) = \prod_{\substack{k \leq 3n \\ 3 \nmid k}} (1 - q^k), \quad c_n(m) = [q^m] P_n(q)^3.$$

P_n^3 is palindromic of degree $9n^2$, so it suffices to consider $m \leq \frac{9}{2}n^2$.

Conjecture 1.1 (cubic Borwein conjecture). For all $n \geq 1$ and $0 \leq m \leq \frac{9}{2}n^2$: $c_n(m) \geq 0$ if $m \equiv 0$, and $c_n(m) \leq 0$ if $m \equiv 1, 2 \pmod{3}$.

The first and second Borwein conjectures [1] concern P_n and P_n^2 ; the first was proved by Wang [4], and both, together with the classes $m \equiv 0, 1$ of Conjecture 1.1, by Wang and Krattenthaler [5]. They explain [5, §11] why the class $m \equiv 2$ resists their method. Writing $P_\infty(q) = \prod_{3 \nmid k} (1 - q^k)$ and

$$E_n(q) = \frac{P_\infty(q)^3}{P_n(q)^3} = \prod_{\substack{k > 3n \\ 3 \nmid k}} (1 - q^k)^{-3},$$

one has $P_n^3 = P_\infty^3 E_n$, and by Jacobi's identity $(q; q)_\infty^3 = \sum_{j \geq 0} (-1)^j (2j+1) q^{j(j+1)/2}$ together with $P_\infty = (q; q)_\infty / (q^3; q^3)_\infty$, all coefficients of P_∞^3 in the class 2 vanish. Hence $c_n(m)$ for $m \equiv 2$ is produced entirely by the tail E_n , and the leading saddle-point contribution cancels.

Theorem 1.2. For all $n \geq 1$ and all $m \equiv 2 \pmod{3}$ with $m \leq \frac{9}{2}n^2$ we have $c_n(m) = 0$ if $m < 3n$ and $c_n(m) < 0$ if $m \geq 3n$. Together with [5] this proves Conjecture 1.1.

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Structure of the proof, and the role of the computer. Put $N = 3n + 1$, $a = \pi^2/9$, $\mu = m/N^2$ and $\tilde{\mu} = (m - N - \frac{1}{4})/N^2$. The admissible pairs (n, m) are covered by five regions (Table 1).

region	method	where
$n \leq 243$	exact integer computation	§3
$n \geq 244, m < 6n + 2$	band lemma	§3.2
$n \geq 244, m \geq 6n + 2, \tilde{\mu} < 0.0985, \text{ saddle } v \leq 3000$	Laplace certificate, uniform in n	§4
$n \geq 244, \text{ saddle } v \geq 3000$	closing lemma	§4.3
$n \geq 244, \mu \geq 0.098$	centre certificate, uniform in n	§5

TABLE 1. The covering of the admissible (n, m) . That these regions exhaust all cases is Proposition 6.1.

The analytic estimates are *not* carried out by hand with numerical constants. Instead, each region is reduced to finitely many inequalities between explicit functions of a few real parameters (a scaled saddle point v or ξ , and $\epsilon = 1/N$), and these are verified on a covering of the parameter domain by balls, in ball arithmetic [3], so that each verified inequality holds for all parameter values in the ball. Every such step is stated below as a *Certificate*, together with the program that checks it and the quantity it outputs. The uniformity in n comes from two devices: the scaled integrands are analytic in $\epsilon = 1/N$ at $\epsilon = 0$, so one ball $\epsilon \in [0, \epsilon_1]$ covers all $N \geq 1/\epsilon_1$; and every term that depends on N is shown (by an asserted inequality) to be non-increasing in N .

The programs, the command that regenerates every log, and the expected outputs are described in §7 and are available as ancillary files [2].

2. STRUCTURAL LEMMAS

Let $\omega = e^{2\pi i/3}$ and $p_3(i) = [q^i]P_\infty(q)^3$.

Lemma 2.1 (tail phase). *For $0 < q < 1$, $\Delta(q) := \text{Im} \log E_n(\omega q) \in (0, \pi/2)$.*

Proof. $\text{Im} \log E_n(\omega q) = 3 \sum_{A \geq n} [f(q^{3A+1}) - f(q^{3A+2})]$ with $f(x) = \arctan(\sqrt{3}x/(2+x))$, increasing on $[0, 1]$ with $f(1) = \pi/6$. Each bracket is positive, and $f(q^{3A+1}) - f(q^{3A+2}) \leq f(q^{3A+1}) - f(q^{3A+4})$ telescopes to at most $f(q^{3n+1}) < \pi/6$. $\square$

Lemma 2.2 (closed forms). *Let $\chi'(r) = 1$ for $r \equiv 1$, -1 for $r \equiv 2$, and $h_r(q) = (q^{Nr} + q^{(N+1)r})/(1 - q^{3r})$. For $|q| < 1$,*

$$\log E_n(\omega q) = S(q) + i\Delta(q), \quad S = - \sum_{3 \nmid r} \frac{3}{2r} h_r + \sum_{3 \mid r} \frac{3}{r} h_r, \quad \Delta = \sum_{3 \nmid r} \frac{3\sqrt{3}}{2r} \chi'(r) \frac{q^{Nr}(1 - q^r)}{1 - q^{3r}},$$

as an identity of analytic functions. Consequently $F(q) := \frac{1}{2i}(E_n(\omega q) - E_n(\bar{\omega} q)) = e^S \sin \Delta$ on $0 < q < 1$, and F is the analytic continuation of this expression.

Proof. Expand $-3 \log(1 - \omega^k q^k) = 3 \sum_r \omega^{kr} q^{kr}/r$ and sum the geometric series over $k \equiv 1, 2 \pmod{3}$, $k > 3n$; this gives $\sum_r \frac{3}{r} (\omega^r q^{Nr} + \omega^{2r} q^{(N+1)r})/(1 - q^{3r})$, whose real and imaginary parts for real q are the displayed expressions, and the displayed combination equals the series identically in q . $\square$

For $i \geq 1$ put $\mathcal{P}_k(i) = \frac{2\pi}{k} \sqrt{B/C} I_1\left(\frac{2\pi}{k} \sqrt{BC}\right)$ with $B = \frac{1}{2}$, $C = 2i - \frac{1}{2}$, and $\mathcal{P}_k(i) = 0$ for $i \leq 0$. Note $\mathcal{P}_3(i) = \sqrt{a/(i - \frac{1}{4})} I_1(2\sqrt{a(i - \frac{1}{4})})$. Let $\chi(i) = 1, -1, 0$ for $i \equiv 0, 1, 2 \pmod{3}$.

Theorem 2.3 (explicit expansion; Appendix A). *For all $i \geq 1$,*

$$p_3(i) = \sqrt{3} \chi(i) \mathcal{P}_3(i) + \rho(i), \quad |\rho(i)| \leq \sum_{\substack{6 \leq k \leq K(i) \\ 3|k}} \varphi(k) \mathcal{P}_k(i) + E_{\text{con}},$$

with $K(i) = \lfloor \sqrt{4\pi i} \rfloor + 2$ and $E_{\text{con}} = 50.1501$.

Lemma 2.4 (Poisson form). *Let $\mathcal{P}(q) = \sum_{i \geq 1} \mathcal{P}_3(i) q^i$. For $\text{Re } w > 0$, $|\text{Im } w| \leq \pi$,*

$$\mathcal{P}(e^{-w}) = e^{-w/4} (e^{a/w} - 1 + \varepsilon(w)), \quad \varepsilon(w) = \sum_{k \neq 0} (-i)^k (e^{a/(w+2\pi i k)} - 1),$$

and $|-1 + \varepsilon(w)| \leq 1.42$.

Proof. $x \mapsto \sqrt{a/x} I_1(2\sqrt{ax})$ has Laplace transform $e^{a/w} - 1$; Poisson summation over the shifted lattice $i - \frac{1}{4}$ gives the identity. The bound is Certificate 2.5. $\square$

Certificate 2.5. `poisson_check.py`: ball-arithmetic enclosure of $|-1 + \varepsilon(w)|$ on 1600 boxes covering $0 < \text{Re } w \leq 10$, $|\text{Im } w| \leq \pi$, with the terms $|k| > 400$ bounded by $8a/(\pi \cdot 400) + a^2 e^{a/\pi}/(400\pi^2)$. Output: $|-1 + \varepsilon| \leq 1.420$.

Proposition 2.6 (reduction). *Let $m \equiv 2 \pmod{3}$, $W > 0$ and $w = W + iy$. Then $c_n(m) = -2 \text{Im } J + R(m)$, where*

$$\text{Im } J = \frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\hat{R}(w)} dy + \text{Im } J_{\varepsilon}, \quad \hat{R}(w) = \frac{a}{w} + \left(m - \frac{1}{4}\right)w + S(e^{-w}) + \log \sin \Delta(e^{-w}),$$

$|\text{Im } J_{\varepsilon}| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} 1.42 e^{(m-\frac{1}{4})W} |F(e^{-w})| dy$, and, for every $w_1 > 0$,

$$|R(m)| \leq \sum_{\substack{6 \leq k \leq K(m) \\ 3|k}} 3\varphi(k) e^{mw_1 + a_k/w_1} E_n(e^{-w_1}) + E_{\text{con}} e^{mw_1} E_n(e^{-w_1}), \quad a_k = \pi^2/k^2.$$

Proof. $c_n(m) = \sum_j e(j) p_3(m-j)$ with $e(j) = [q^j] E_n \geq 0$. Inserting Theorem 2.3, the χ -part equals $\sum_j e(j) \sqrt{3} \chi(m-j) \mathcal{P}_3(m-j) = -2 \text{Im}[q^m] \mathcal{P}(q) E_n(\omega q)$ for $m \equiv 2$ (compare coefficients: $e(j)$ is real and $\sqrt{3} \chi(m-j) = -2 \text{Im } \omega^j$ for $m \equiv 2$). The coefficient is $\frac{1}{2\pi} \int \mathcal{P}(e^{-w}) E_n(\omega e^{-w}) e^{mw} dy$; by Lemma 2.4, reality of $[q^m] \mathcal{P} F$ and Lemma 2.2 this gives the displayed $\text{Im } J$. For R , the remainder terms are bounded termwise using $e(j) \geq 0$, $\mathcal{P}_k \geq 0$ and $\sum_i \mathcal{P}_k(i) e^{-w_1 i} \leq (1 + 2a_k) e^{a_k/w_1} \leq 3e^{a_k/w_1}$ (from $I_1(y) \leq \frac{y}{2} e^y$ and the Laplace transform). $\square$

Remark 2.7. Proposition 2.6 is checked end to end by `identity_check.py`: for $n = 12$ ($m = 200, 302, 404$) and $n = 20$ ($m = 500, 800, 1100$) the exact $c_n(m)$ agrees with $-2(\text{Im } J_0 + \text{Im } J_{\varepsilon}) + R(m)$, each piece computed independently, to relative error $< 10^{-8}$. The bound of Theorem 2.3 is checked exactly for all $i \leq 3000$ and for 3388 values $i \leq 108000$ (`expansion_check.py`); the maximal ratio is $\sqrt{3}/2$.

In particular, $c_n(m) < 0$ follows from

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\hat{R}(W+iy)} dy > |\text{Im } J_{\varepsilon}| + \frac{1}{2} |R(m)|. \quad (1)$$

3. THE FINITE RANGE AND THE BAND

Certificate 3.1 (finite range). `exact_small_n.py` computes P_n^3 exactly in $\mathbb{Z}[q]$ (FLINT) for $n \leq 243$ and checks $c_n(m) = 0$ for $m < 3n$ and $c_n(m) < 0$ for $3n \leq m \leq \frac{9}{2}n^2$, $m \equiv 2$. Output: no violations.

Proposition 3.2 (band). For $3n + 2 \leq m < 6n + 2$ with $m = 3M + 2$, $c_n(m) = 3 \sum_{i=0}^{M-n} g(i)$ with $g(i) = p_3(3i) + p_3(3i + 1)$, and $g(i) < 0$ for all $i \geq 0$. Hence $c_n(m) < 0$ in the band.

Proof. Since $6n + 2 = 2N$, $E_n \equiv 1 + 3 \sum_{N \leq j < 2N, 3 \nmid j} q^j \pmod{q^{2N}}$, and $p_3(m) = 0$. Grouping $j = 3A + 1$ and $j = 3A + 2$ gives the identity (checked exactly for $n \leq 60$ by `band_lemma.py`). Negativity of g is exact for $i \leq 36000$ (Certificate 3.3). For $i \geq 36000$, Theorem 2.3 gives $g(i) \leq -\sqrt{3}(\mathcal{P}_3(3i + 1) - \mathcal{P}_3(3i)) + \rho_{\max}(3i) + \rho_{\max}(3i + 1)$. Since $\mathcal{P}_3(x) = 2aI_1(y)/y$ with $y = 2\sqrt{a(x - \frac{1}{4})}$, we have $\mathcal{P}_3'(x) = 4a^2I_2(y)/y^2$, and $\mathcal{P}_3$ is increasing and convex, so the difference is at least $\mathcal{P}_3'(3i)$. Using $I_2(y) \geq \kappa_2 e^y / \sqrt{2\pi y}$ for $y \geq y_0$, $I_1(y) \leq e^y / \sqrt{2\pi y}$, $\mathcal{P}_k \leq \mathcal{P}_6$ for $k \geq 6$ and $\sum \varphi(k) \leq K^2/2$, the ratio of the ρ -bound to the main difference is of the form $\text{poly}(y)e^{-y/2}$, decreasing for $y \geq 9$, and at $i = 36000$ ($y_0 = 688.3$) it is at most $5 \cdot 10^{-139}$ in ball arithmetic (`band_tail_rigorous.py`). $\square$

Certificate 3.3. `band_lemma.py`: exact p_3 from Jacobi's identity and the recurrence $jb(j) = 3 \sum_t \sigma(t)b(j-t)$ for $b = [q^j](q; q)_\infty^{-3}$. Output: identity holds for $n \leq 60$; $g(i) < 0$ for all $i \leq 36000$.

4. THE LAPLACE REGIME

4.1. Scaled form. Let $\zeta = Nw$, $\epsilon = 1/N$, $g(x) = x/(1 - e^{-x})$ and $h(x) = (1 - e^{-x})/(1 - e^{-3x}) = g(3x)/(3g(x))$. Both g and h are analytic for $|x| < 2\pi/3$.

Lemma 4.1. $S(e^{-w}) = N\tilde{S}(\zeta, \epsilon)$ and $\hat{R}(w) = NG(\zeta) + (m - N - \frac{1}{4})\zeta/N$, where

$$\tilde{S} = \sum_r s_r e^{-r\zeta} (1 + e^{-r\zeta\epsilon}) \frac{g(3r\zeta\epsilon)}{3r\zeta}, \quad \tilde{Q} = \sum_{3 \nmid r} \frac{3\sqrt{3}}{2r} \chi'(r) e^{-(r-1)\zeta} h(r\zeta\epsilon),$$

$s_r = -\frac{3}{2r} (3 \nmid r)$, $\frac{3}{r} (3 \mid r)$, and $G = \frac{a}{\zeta} + \tilde{S} + \epsilon(\log \tilde{Q} + \log \frac{\sin \Delta}{\Delta})$ with $\Delta = \tilde{Q}e^{-\zeta}$. The saddle condition $\hat{R}'(W) = 0$ reads $-G'(v) = \tilde{\mu}$ with $v = NW$.

Proof. Substitute $q = e^{-\zeta\epsilon}$ in Lemma 2.2; $\log \sin \Delta = \log \tilde{Q} - \zeta + \log \frac{\sin \Delta}{\Delta}$, and the $-\zeta$ combines with $(m - \frac{1}{4})w$ to $(m - N - \frac{1}{4})\zeta/N$. $\square$

Thus a pair (N, m) in this regime is represented by its real saddle v and $\epsilon = 1/N$. Since $m \geq 2N$, $\tilde{\mu} \geq \epsilon(1 - \frac{1}{4N}) \geq 0.999\epsilon$; balls with $-G'(v) < 0.999\epsilon$ on the whole ball contain no pairs.

4.2. The error budget on a ball. Fix a ball $V \ni v$ and a ball $E \ni \epsilon$ with $E \subset [0, 1/733]$, and let $N_b = \max(733, 1/\sup E)$. Write $G_j = G^{(j)}(v)$, $\sigma_y = N^{-3/2}G_2^{-1/2}$ and $\tau = y/\sigma_y$. Dividing (1) by $e^{\hat{R}(W)}\sigma_y/(2\pi)$, it suffices to show

$$B := \frac{Z_1 + Z_2 + Z_3 + Z_\epsilon + Z_R}{L} < 1, \quad (2)$$

where L bounds the window integral from below and the Z 's bound the remaining contributions from above:

Window $|\tau| \leq c$. $\hat{R}(W + i\sigma_y\tau) - \hat{R}(W) = -\frac{\tau^2}{2} - iG_{3,n}\frac{\tau^3}{6} + \phi_4\frac{\tau^4}{24} + E_5$ with $G_{3,n} = \sqrt{\epsilon}|G_3|G_2^{-3/2}$, $\phi_4 = \epsilon G_4 G_2^{-2}$, $|E_5| \leq G_{5,n}|\tau|^5/120$, $G_{5,n} = \epsilon^{3/2}M_5 G_2^{-5/2}$, where M_5 bounds $|G^{(5)}|$ on the window: $120a/v^6$ for the a/ζ part plus a Cauchy estimate (Lemma B.1) for the rest on the

circle $|\zeta - v| = 0.6v$. $L = \int_{-c}^c \ell(\tau) d\tau$ with ℓ the resulting pointwise lower bound of the real part ($\cos \theta \geq 1 - \theta^2/2$).

Zone 1, $c \leq |\tau|$, $|t| \leq t_s$ ($t = Ny$). $\operatorname{Re}[G(v + it) - G(v)] \leq -\frac{t^2}{2}(\kappa_0 - \epsilon\kappa_b)$ from the even Taylor terms of $G_0 = a/\zeta + \tilde{S}$ and of $b = \log \tilde{Q} + \log \frac{\sin \Delta}{\Delta}$ with Cauchy remainders at order 6; thus $Z_1 = 2 \int_c^\infty e^{-\kappa\tau^2/2} d\tau$, $\kappa = (\kappa_0 - \epsilon\kappa_b)/G_2$, independent of N .

Zone 2, $t_s \leq t \leq t_C$. On cells, $\operatorname{Re} \hat{R} - \hat{R}(W) = NA(t) + B(t)$ with $A = \operatorname{Re}[G_0(v + it) - G_0(v)]$ and $B = \operatorname{Re}[b(v + it) - b(v)]$, enclosed in ball arithmetic (the a/ζ part in closed form, the factor e^{-rv} separated to control dependency). The certificate asserts $A < -1/(2N_b)$ on every cell, so $N^{1/2}e^{NA} \leq N_b^{1/2}e^{N_b A}$ for $N \geq N_b$, and $Z_2 = (N_b G_2)^{1/2} \sum_{\text{cells}} 2(e^{N_b A + B} + 3e^{N_b(A - \operatorname{Re}(a/\zeta)) + B}) |\text{cell}|$ (the second term is the ε -part).

Zone 3, $t \geq t_C$. Either (crude) $|F| \leq E_n(e^{-W})$ and $|e^{a/w} - 1 + \varepsilon| \leq e^{aW/(W^2 + y^2)} + 3$, giving an exponent $N\beta + O(1)$ with $\beta = \Psi(v)/v + av/(v^2 + t_C^2) - a/v - \tilde{S}(v)$, $\Psi(v) = 2e^{-v}/(1 - e^{-v})$, asserted $\beta < -3/(2N_b)$; or (split, when $\inf E > 0$) $|\tilde{Q}| \leq \sum_{3\uparrow r} \frac{3\sqrt{3}}{2r} e^{-(r-1)v} 2(1/(3rv \inf E) + 1)$ and $|\tilde{S}|$ bounded termwise, with slope $< -5/(2N_b)$.

$R(m)$. Proposition 2.6 with $w_1 = t'W$, t' from a fixed list, $\log E_n(e^{-w_1}) \leq N\Psi(Nw_1)/(Nw_1) + 6|\log(1 - e^{-Nw_1})|$, $\tilde{\mu} = -G'(v)$ and φ -sum $\leq 3(K^2/6 + K)$; each exponent is asserted to have slope $< -7/(2N_b)$.

Lemma 4.2. *If all assertions above hold on (V, E) , then B computed with $N = N_b$ bounds (2) for every N with $1/N \in E$ and every m whose saddle lies in V .*

Proof. L and Z_1 do not depend on N (their N -dependence enters only through $\epsilon \in E$). Every other term is of the form $N^p e^{N\lambda}$ with λ bounded by the asserted slope $< -p/N_b$, hence non-increasing in $N \geq N_b$. $\square$

Certificate 4.3 (uniform Laplace grid). `run_uniform.py` covers $v \in [2.95, 3000]$ by 36 chunks of balls and, for each v -ball, $\epsilon \in [0, 1/733]$ by ϵ -balls; balls whose pairs are vacuous are discarded only after the rigorous bound $\tilde{\mu} \leq a/v^2 + \max_{|\zeta - v| = v/2} |h(\zeta) - h(v)|/(v/2)$ (with $h = \tilde{S} + \epsilon b$) proves $\tilde{\mu} < 0.999\epsilon$, or when $\tilde{\mu} > 0.0985$ (centre regime). Failing balls are bisected. Output: every chunk certified, 0 unresolved balls, $\max B = 0.117$.

4.3. The closing lemma.

Proposition 4.4. *For every pair with saddle $v \geq 3000$, (2) holds with $B \leq 0.0082$.*

Proof and certificate. Put $s = \epsilon v^2/a$; the constraint $\tilde{\mu} \geq 0.999\epsilon$ and $\tilde{\mu} = (a/v^2)(1 + O(e^{-v}))$ give $s \leq 1.0011$. On $|\zeta - v| \leq v/2$: $|\tilde{S}| \leq \sum_r \frac{3}{r} e^{-rv/2} 2(\frac{2}{3rv} + \epsilon) \leq 5 \cdot 10^{-655}$; $|h(x)| \leq 4$ for every r (by the Bernoulli enclosure if $|x| < 0.9$, and $|h(x)| \leq 2(|x|/(3 \operatorname{Re} x) + 1)$ with $|x|/\operatorname{Re} x = |\zeta|/\operatorname{Re} \zeta \leq 3$ otherwise), so the $r \geq 2$ part of $\tilde{Q}$ and Δ are $\leq 4 \cdot 10^{-651}$; and $\log h$ on the disc $|x| \leq \frac{3}{2}as/v$ is enclosed, giving $|b(\zeta) - b(v)| \leq 4.4 \cdot 10^{-3}$. By Lemma B.1, the relative perturbations of $G_2, \dots, G_5$ from their pure a/ζ values are $\leq 4.7 \cdot 10^{-5}$, so $G_{3,n} \leq 0.0388$, $\phi_4 \leq 2.0 \cdot 10^{-3}$, $G_{5,n} \leq 1.03 \cdot 10^{-3}$, $\kappa \geq 0.8(1 - 10^{-5})$ on $|t| \leq v/2$, and $L \geq 2.494$, $Z_1 \leq 0.0205$. For $|t| \geq v/2$ the exponent is at most $-\frac{v}{5s}(1 - 10^{-5}) + \log(\frac{4v}{3as} + 6) + 20ve^{-v}/(as)$ with prefactor $\sqrt{2}v^{3/2}/(as^{3/2})$; its s -derivative is positive for $s \leq 1.0011$ because $v/5 > 2.5 \cdot 1.0011 + 20ve^{-v}/a$, and it decreases in v , so the worst case is $v = 3000$, $s = 1.0011$, where $Z_2 \leq 5 \cdot 10^{-251}$. The R -term is $\leq 10^{-1278}$. All numbers are ball enclosures (`ray_asymptotic2.py`). $\square$

5. THE CENTRE REGIME

Here $c_n(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_n(re^{i\theta})^3 r^{-m} e^{-im\theta} d\theta$ is treated directly, without Proposition 2.6. Put $L(u) = 3 \sum_{k \leq 3n, 3 \nmid k} \log(1 - e^{ku})$, $u_0 = 2\pi i/3$, and let $u^* = u_0 + x^*$ be the saddle $L'(u^*) = m$ near u_0 ; write $\xi = -N \operatorname{Re} x^*$ and $\eta = N \operatorname{Im} x^*$. The conjugate saddle near $-u_0$ contributes the complex conjugate.

Lemma 5.1 (sign margin). *The phase of the main term is $\alpha(r) - \frac{2\pi}{3} + b + E_1 \pmod{2\pi}$, where $\alpha(r) = -\frac{\pi}{6} + \arctan(\sqrt{3}r^{3n}/(2 + r^{3n}))$, $b = -\frac{1}{2} \arg L''(u^*)$, and $|E_1| \leq \sup |L''| \delta^2/2$ with $\delta = \operatorname{Im} x^*$; moreover $|\delta| \leq 3N \operatorname{TV}_s / (N^3 \rho_{10} - 3N^2 V)$ by the saddle equation and Abel summation.*

Proof. The main term is $M = e^{L(u^*) - mu^*} \omega e^{ib} \sqrt{2\pi/|L''(u^*)|}/(2\pi)$ (steepest descent through u^* in the direction ie^{ib}). Write $u'_0 = \log r + 2\pi i/3$, so $u^* = u'_0 + i\delta$. Then $\operatorname{Im}[L(u^*) - mu^*] = \operatorname{Im} L(u'_0) + \int_0^\delta (\operatorname{Re} L'(u'_0 + is) - m) ds - m \cdot \frac{2\pi}{3}$, and $\operatorname{Re} L'(u^*) = m$ gives $|\operatorname{Re} L'(u'_0 + is) - m| \leq \sup |L''| |\delta - s|$, whence the bound on E_1 . Next $\operatorname{Im} L(u'_0) = \arg P_n(r\omega)^3 = \alpha(r)$ by the pairing $(1 - r^{3j+1}\omega)(1 - r^{3j+2}\omega^2)$ of Lemma 2.1 type; $m \equiv 2$ gives $m \cdot \frac{2\pi}{3} \equiv \frac{4\pi}{3}$, and $\arg \omega = \frac{2\pi}{3}$. For δ : $\operatorname{Im} L'(u'_0) = 3 \sum_i [(3i+2)s(x_{3i+2}) - (3i+1)s(x_{3i+1})]$ with $s(x) = \operatorname{Im}(\omega x/(1 - \omega x))$ and $x_k = r^k$, which is at most $3N \operatorname{TV}_\tau(\tau s(e^{-\xi\tau}))$ in modulus; and $\operatorname{Im} L'(u'_0 + i\delta) = 0$ with $\operatorname{Re} L'' \geq N^3 \rho_{10} - 3N^2 V$ on the segment. $\square$

The certificate splits the circle into a window $|\theta - \theta^*| \leq c\sigma$, a middle stretch up to $|\theta - 2\pi/3| \leq 0.05$, and the far region. Each contribution is bounded uniformly in $N \geq 733$ as a function of ξ :

- *Window*: the cubic Taylor constant $G_3 = H_3\sigma^3$ with $H_3(\text{window}) \leq N^4[h_3 + 3(\operatorname{TV} + S)/N]$ and $\sigma^3 \leq (N^3 a_2 - 3N^2 V)^{-3/2}$ (continuum integrals plus Koksma remainders, Lemma B.3); the window shrinks with N .
- *Middle*: three zones in $t = N(\theta - 2\pi/3)$: Taylor with $\operatorname{Re} L'' \geq N^3 \rho_{10}(t) - 3N^2 V$ ($|t| \leq 0.8$); the linearised continuum profile $-N \int_0^1 g(e^{-\xi\tau})(1 - \cos \tau t) d\tau$ plus a Koksma constant ($0.8 \leq t \leq 10$, slope asserted); a Dirichlet-kernel bound ($t \geq 10$, slope asserted).
- *Far*: for $\theta \geq 0.05$ a Dirichlet-kernel bound; near $\theta = 0$ a continuum profile $-N \int g(1 + 2 \cos \tau t)$ with positivity asserted on $t \in [0, 40]$ and a kernel bound beyond; slopes asserted.
- *Sign margin*: Lemma 5.1 with $r^{3n} = e^{-\xi(1-1/N)}$ enclosed over all $N \geq 733$.

Certificate 5.2 (centre tables). `centre_mid_uniform*.py`, `centre_g3_uniform*.py`, `centre_clow_uniform.py` produce, on 110 ξ -balls covering $[-0.3, 3.4]$, uniform bounds for the middle stretch, the window constant, the sign margin (with $\cos < 0$ asserted) and the far region. `coverage.py` (check E) combines them into $\text{err} = (2(\text{inner} + \text{middle}) + \text{far})/(2|\cos|)$ on every ball. Output: $\text{maxerr} = 0.49 < 1$. The saddle offset satisfies $|\eta| \leq 0.0155$, within the shift margins of the tables (check G).

6. COVERAGE

Proposition 6.1. *Every (n, m) with $n \geq 244$, $m \equiv 2 \pmod{3}$, $6n + 2 \leq m \leq \frac{9}{2}n^2$ lies in the Laplace regime with saddle in $[2.95, 3000]$ or $[3000, \infty)$, or in the centre regime with $\xi \in [-0.3, 3.4]$.*

Proof. (A) the chunks of Certificate 4.3 tile $[2.95, 3000]$ with no unresolved balls. (B) $-G'(2.95) \geq 0.1015 > 0.0985$ for all $\epsilon \in [0, 1/733]$ and $G'' > 0$, so every pair with $\tilde{\mu} < 0.0985$ has $v > 2.95$. (C) $m \geq 2N$ gives $\tilde{\mu} \geq 0.999\epsilon$. (D) $\tilde{\mu} \geq 0.0985$ implies $\mu > \tilde{\mu} \geq 0.098$. (F) $\mu_N(\xi) = N^{-2} \operatorname{Re} L'(u^*)$ is decreasing in ξ and, by Lemma B.3, $\mu_N(3.4) \leq 0.0886 < 0.098$ and $\mu_N(-0.3) \geq 0.531 > \frac{1}{2}$ for all $N \geq 733$, while $m \leq \frac{9}{2}n^2$ gives $\mu < \frac{1}{2}$. Each item is checked by `coverage.py`. $\square$

Proof of Theorem 1.2. Certificate 3.1 for $n \leq 243$; for $n \geq 244$ Proposition 3.2 for $m < 6n + 2$, and otherwise Proposition 6.1 with Lemma 4.2, Certificate 4.3, Proposition 4.4 and Certificate 5.2. $\square$

7. COMPUTATIONS AND REPRODUCIBILITY

All certificates use Arb ball arithmetic through `python-flint 0.9.0` [3]; auxiliary checks use `mpmath 1.3.0`, `numpy 2.4.4`, `scipy 1.17.1` and `sympy 1.14.0`. The single command `./run_all.sh` regenerates every table and log from scratch (about three hours on one core; the uniform Laplace grid parallelises over chunks). The accompanying README lists, for each certificate, the program, the log it writes, and the line that must appear. Independent consistency checks that are not part of the proof are also included: the reduction identity (`identity_check.py`), the exact expansion bound up to $i = 108000$, randomised checks of Lemmas B.1–B.3, and a comparison of both main terms with exact coefficients at $n = 244$ (agreement to 0.1–0.7%).

Load-bearing hand arguments. The proof depends on the following written arguments in addition to the certificates: Lemmas 2.1–2.4, Proposition 2.6, Theorem 2.3 (Appendix A), Lemma 4.1, the derivation of the budget (2), Lemma 4.2, the analytic parts of Proposition 4.4, Lemma 5.1, and Appendix B.

APPENDIX A. PROOF OF THEOREM 2.3

We specialise the circle method for $G_p(q)^\delta$, $G_p = \prod_{m \geq 1} (1 - q^m)/(1 - q^{pm})$, to $p = \delta = 3$; then $G_3^3 = P_\infty^3$. Let $\tau = h/k + iz/k^2$, $d = \gcd(3, k)$, and $f(x) = \prod_{m \geq 1} (1 - x^m)^{-1}$. The eta transformation gives

$$G_3(e^{2\pi i \tau})^3 = \left(\frac{3}{d}\right)^{3/2} e^{3\pi i(-s(h,k)+s(3h/d,k/d))} \exp\left(\frac{\pi(d^2-3)}{12z} - \frac{\pi z}{2k^2}\right) \widehat{G}^3, \quad \widehat{G} = \frac{f(e^{2\pi i d h'_d/k - 2\pi d^2/(3z)})}{f(e^{2\pi i h'/k - 2\pi/z})},$$

with $hh' \equiv -1 \pmod{k}$ and principal powers.

Growing and bounded cusps. If $3 \mid k$ then $d = 3$, the prefactor has modulus 1 and the exponential is $e^{\pi/(2z)}$: every such cusp grows with $B = \frac{1}{2}$, and the correction term of $\widehat{G}^3$ grows only if $\delta > 24/(p-1) = 12$, which is not the case. If $3 \nmid k$, then on $\operatorname{Re}(1/z) = w \geq 1$, $|G_3^3| \leq K_3^3$ with $K_3 = \sqrt{3} e^{-\pi/18} f(e^{-2\pi/3}) f(e^{-2\pi}) = 1.691256$, the maximum being attained at $w = 1$.

Rademacher dissection. Take Farey order $N' \geq \sqrt{4\pi i} + 1$. On the Ford-circle arcs mapped to $K = \{|z - \frac{1}{2}| = \frac{1}{2}\}$, the endpoints satisfy $|z'|, |z''| \leq \sqrt{2k}/(N' + 1)$ and $\operatorname{Re} z \leq 2k^2/(N' + 1)^2$, so $|e^{\pi C z/k^2}| \leq e$ with $C = 2i - \frac{1}{2}$.

Main terms. For $3 \mid k$, the term 1 of $\widehat{G}^3$ gives $\exp(\pi/(2z) + \pi C z/k^2)$; completing the arc to K and using $\frac{i}{k^2} \int_K \exp(\pi B/z + \pi C z/k^2) dz = \mathcal{P}_k(i)$ yields the main term with amplitude of modulus 1, for each of the $\varphi(k)$ values of h . The completion costs at most $\frac{1}{k^2} \cdot 2 \cdot \frac{\pi}{2} \cdot \frac{\sqrt{2k}}{N'+1} e^{1+\pi/2}$ per cusp, and $\sum_{3 \mid k \leq N'} \varphi(k)/k \leq \frac{2}{9}(N' + 1)$ because $\varphi(3j)/(3j) \leq \frac{2}{3}$; total $E_{\text{arc}} = \sqrt{2\pi} \frac{2}{9} e^{1+\pi/2} = 12.9103$.

Remainder at growing cusps. Moving to the chord (length $\leq 2\sqrt{2k}/(N' + 1)$), $|e^{\pi/(2z)}(\widehat{G}^3 - 1)| \leq e^{\pi w/2} (f(e^{-2\pi w})^3 f(e^{-6\pi w})^3 - 1)$, whose terms $e^{\pi w/2} e^{-2\pi j w}$ ($j \geq 1$) are non-increasing, so the maximum is at $w = 1$; total $E_{\text{rem}} = 2\sqrt{2} \frac{2}{9} e^{1+\pi/2} (f(t)^3 f(t^3)^3 - 1) = 0.0463$, $t = e^{-2\pi}$.

Bounded cusps. On the chords, the integrand is at most eK_3^3 ; summing $\frac{1}{k^2} \frac{2\sqrt{2k}}{N'+1}$ over $\leq \varphi(k) \leq k$ values of h and $k \leq N'$ gives $E_{\text{ng}} = 2\sqrt{2} e K_3^3 = 37.1935$.

The $k = 3$ amplitude. For $k = 3$ we have $d = 3$, $(3/d)^{3/2} = 1$ and $s(3h/d, k/d) = s(h, 1) = 0$, so the phase of the cusp $h/3$ is $e^{-3\pi i s(h,3)}$. The Dedekind sums are $s(1, 3) = \frac{1}{18}$ and $s(2, 3) = -\frac{1}{18}$,

and the leading term of $\widehat{G}^3$ is 1. Hence the two cusps contribute $\mathcal{P}_3(i)$ times

$$e\left(-\frac{i}{3}\right)e^{-\pi i/6} + e\left(-\frac{2i}{3}\right)e^{\pi i/6} = 2\cos\left(\frac{2\pi i}{3} + \frac{\pi}{6}\right) = \sqrt{3}, -\sqrt{3}, 0 \quad (i \equiv 0, 1, 2 \pmod{3}),$$

that is, $\sqrt{3}\chi(i)\mathcal{P}_3(i)$. (This is also confirmed by the exact agreement recorded in the remark after Proposition 2.6.) For $k \geq 6$ only the modulus 1 of each amplitude is used. Summing, $E_{\text{con}} = E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}} = 50.1501$. The constants are recomputed by `sz.expansion.audit.py`. $\square$

APPENDIX B. ANALYTIC TOOLS

Lemma B.1 (Cauchy). *If f is holomorphic on $|\zeta - c| \leq \rho$ and $|f - f(c)| \leq M$ on $|\zeta - c| = \rho$, then $|f^{(k)}(p)| \leq k!M\rho/(\rho - d)^{k+1}$ for $|p - c| \leq d < \rho$, $k \geq 1$.*

Proof. $f^{(k)}(p) = \frac{k!}{2\pi i} \oint (f(\zeta) - f(c))(\zeta - p)^{-k-1} d\zeta$ and $|\zeta - p| \geq \rho - d$. $\square$

Lemma B.2 (midpoint rule). *For $g \in C^2[-\frac{1}{2}, \frac{1}{2}]$, $|g(0) - \int g| \leq \frac{1}{8} \int |g''|$.*

Proof. $g(0) - \int g = -\int Kg''$ with $K(t) = \frac{1}{2}(\frac{1}{2} - |t|)^2 \leq \frac{1}{8}$. $\square$

Lemma B.3 (Koksma, one node per cell). *If $x_i \in [(i-1)h, ih]$ for $i = 1, \dots, M$, then $|h \sum f(x_i) - \int_0^1 f| \leq h \text{TV}(f) + (1 - Mh) \sup |f|$.*

Proof. $|hf(x_i) - \int_{C_i} f| \leq h \text{TV}_{C_i}(f)$; sum, and bound the uncovered part. $\square$

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DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY, WEST LAFAYETTE, IN 47907, USA
 Email address: `jpadhi@purdue.edu`